

AI SMART FARMING ADVICE AGENT

INTRODUCTION

Agriculture plays a vital role in ensuring food security and supporting the economy. However, farmers often face challenges such as changing weather conditions, pest attacks, soil degradation, improper irrigation, and fluctuating market prices. Access to reliable and timely agricultural information is limited, making it difficult to take informed farming decisions. An AI-powered farming assistant can help farmers improve productivity and adopt sustainable farming practices.

PROBLEM STATEMENT

Modern agriculture faces several challenges, including unpredictable weather, pest and disease outbreaks, declining soil fertility, and changing market conditions. Farmers frequently depend on scattered or unreliable sources of information, which can lead to poor crop management and reduced profitability. There is a need for an intelligent system that provides personalized, accurate, and real-time farming guidance to support better agricultural decision-making.

OBJECTIVES

The main objectives of the Smart Farming Advice Agent are:

- Develop an AI-powered farming assistant using IBM watsonx.ai.
- Recommend suitable crops based on soil type, weather, season, and location.
- Provide irrigation, fertilizer, and pest management guidance.
- Retrieve trusted agricultural knowledge using Retrieval-Augmented Generation (RAG).
- Display weather forecasts and market price information.
- Generate personalized farming recommendations through an interactive dashboard

PROPOSED SOLUTION

The proposed solution is an AI-powered Smart Farming Advice Agent developed using **Python Flask**, **IBM watsonx.ai**, **IBM Granite-3-8B-Instruct**, and **Retrieval-Augmented Generation (RAG)**. The system retrieves trusted agricultural information from a knowledge base, combines it with weather forecasts, market prices, and farm profile details, and generates personalized recommendations for farmers. It also provides an interactive dashboard, AI-powered chat interface, crop recommendations, and farming reports.

TECHNOLOGIES USED

- Python Flask
- IBM watsonx.ai
- IBM Granite-3-8B-Instruct
- IBM Cloud
- Retrieval-Augmented Generation (RAG)
- FAISS Vector Database
- Weather API
- Market Price API
- HTML, CSS, JavaScript

KEYWORDS

- AI-powered farming assistant
- Personalized crop recommendations
- Weather forecasting and advisories
- Fertilizer recommendations
- Irrigation management
- Pest and disease guidance
- Market price updates
- Farm profile management
- Interactive dashboard
- AI-generated farming reports

EXPECTED OUTCOME

The Smart Farming Advice Agent helps farmers make informed decisions by providing accurate and personalized agricultural recommendations. The system improves crop productivity, reduces farming risks, supports sustainable farming practices, and enhances overall farm management through AI-powered guidance.

FUTURE SCOPE

- Integration with IoT-based smart farming sensors.
- AI-powered voice assistant with multilingual support.
- Satellite and drone image analysis for crop monitoring.
- AI-based crop disease detection using images.
- Advanced predictive analytics for crop yield estimation.

CONCLUSION

The AI Smart Farming Advice Agent combines artificial intelligence, IBM watsonx.ai, IBM Granite Models, and RAG to deliver reliable and personalized farming guidance. The solution enables farmers to make data-driven decisions, improve agricultural productivity, reduce risks, and promote sustainable farming practices.

